

Ingestion Report: RNA-seq Data Processing and Cleaning

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For my ingestion, I focused on cleaning and preparing two RNA-seq datasets from the Gene Expression Omnibus (GEO): GSE152418 and GSE156063. The main goal was to take the raw gene expression data and messy metadata files and turn them into clean, structured datasets that are ready for analysis.

I first worked on the GSE152418 dataset in the Cleaning Comp Dataset. I loaded in the raw gene count data, gene annotations, and the GEO series matrix file. The metadata in the GEO file was not in a clean format, so I created a helper function to extract specific rows that contain useful information like sample IDs, titles, days post symptom onset, gender, disease status, and severity. I then combined all of these into one metadata dataframe. This made it much easier to work with compared to the original format, where everything was embedded in text. I also checked the structure and dimensions of the datasets to make sure it was lined up correctly.

Next, I worked on the GSE156063 dataset in Cleaning Data RNA, which needed more detailed cleaning. I started by loading the gene count data and metadata file. The metadata had a lot of formatting issues, so I removed tabs and quotation marks and then split it into key value pairs. After that, I reshaped the data from a long format into a wide format so that each sample has its own row and each variable has its own column.

For the gene expression data, I cleaned the count matrix by renaming the gene column, converting all values to numeric, and transposing the data so that samples are rows and genes are columns. I then did a log transformation using `log1p()` to make the data less skewed and easier to

analyze. After transforming it, I turned it back into a dataframe and added sample IDs so it matches the metadata.

I also cleaned up the GEO series matrix file by removing unnecessary sections and keeping only the metadata portion before the expression table starts. Then I worked on organizing the sample characteristics into separate variables so that they are easier to use later.

Across both datasets, I made sure to rename columns so they are more clear and consistent, removed any unnecessary variables, and made sure it is aligned by sample ID. This makes the datasets much easier to understand and use for future analysis.

Overall, this ingestion process took raw and messy RNA-seq data and turned it into clean, structured datasets that are ready for further analysis like visualization or differential expression using tools like DESeq2. Thank you.